

ERMANNNO BARTOLI

Researcher in Robot Learning for HRI

@ bartoli@kth.se ermannno-bartoli Ermanno-Bartoli Publications



EXPERIENCE

Ph.D. Researcher

KTH Royal Institute of Technology

Dec 2022 - Present Stockholm, Sweden

- 3D Scene Graph generation
- Semantic Scene Understanding
- Continual Learning for HRI

Research Engineer

KTH Royal Institute of Technology

Jul 2022 - Sep 2022 Stockholm, Sweden

- Joined joined SoRo lab as a Research Engineer
- Working on social motion planning

Researcher

Sapienza University of Rome

Jan 2020 - Jul 2020 Rome, Italy

- Joined Lab RoCoCo (Cognitive Cooperating Robots)
- Working on Planning for Humanoid Robots for RoboCup Competition

SELECTED PROJECTS

Real-time RRT* with Signal Temporal Logic Preferences
We developed a method to integrate Signal Temporal Logic (STL) constraints into the Real-Time Rapidly Exploring Random Tree (RT-RRT*) motion planning algorithm for dynamic environments, guiding the algorithm towards solutions that maximally adhere to STL specifications for **social navigation** scenarios.

Winner of Episode 2, SciRoc Challenge
I was the **Team Leader** of Sapienza team “SPQR CocoNuts”, which participated in SciRoc competition 2021 where we used TIAGo Robot to recognize and generate signs belonging to the Italian Sign Language.

HOBBIES

Football, Tennis

Chess

REFERENCE

- Iolanda Leite, iolanda@kth.se
- Daniele Nardi, nardi@dis.uniroma1.it

RESEARCH

Continual Learning for HRI in real-world applications

My research is centered Human-centered scene understanding and long-term robot adaptation for dynamic home environments.

EDUCATION

Doctorate in Robotics

KTH Royal Institute of Science

2022.12 – present Sweden

Master in AI & Robotics

Sapienza University of Rome

2020.10 – 2022.07 Italy
Grade: 110/110 with honors

Bachelors in Computer Engineering

Sapienza University of Rome

2017.10 – 2020.07 Italy
Grade: 110/110 with honors

PROGRAMMING

Python
ROS
C/C++
Dart



STRENGTHS

Continual Learning Research
Robot Programming Deep Learning
Project Management

LANGUAGES

English
Italian
Swedish
Spanish



PUBLICATION LIST

3D Scene Graph generation & Semantic Scene Understanding

1. **Bartoli**, et al. "Long-Term Planning Around Humans in Domestic Environments with 3D Scene Graphs". Workshop on Lifelong Learning and Personalization in Long-Term Human-Robot Interaction (LEAP-HRI), ACM/IEEE International Conference of Human-Robot Interaction (HRI) 2025.
2. **Bartoli**, et al. "Social 3D Scene Graphs: Modeling Human Actions and Relations for Interactive Service Robots". Under Submission to IEEE International Conference on Robotics and Automation (ICRA) 2026.

Continual Learning

1. **Bartoli**, et al. "Knowledge acquisition and completion for long-term human-robot interactions using knowledge graph embedding." International Conference of the Italian Association for Artificial Intelligence, 2022.
2. **Bartoli**, et al. "SAdaGR: Semantically Adaptive Generative Replay for Continual Learning." In preparation.
3. **Bartoli**, et al. "STREAK: Streaming Network for Continual Learning of Object Relocations under Household Context Drifts" IEEE International Conference on Robot and Human Interactive Communication (RO-MAN) 2025.

Human-Robot Interaction

1. **Bartoli**, et al. "The Need for (Robot) Speed: Offloading Heavy Computations Improves Response Time and User Experience in Spoken Interactions" IEEE International Conference on Robot and Human Interactive Communication (RO-MAN) 2025.
2. Antonioni,(...),**Bartoli** et al. "Nothing about us without us: A participatory design for an inclusive signing Tiago robot." IEEE International Conference on Robot and Human Interactive Communication (RO-MAN) 2022.
3. Torre, (...), **Bartoli** et al. "Real-Time RRT* with Signal Temporal Logic Preferences". IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS) 2023.
4. **Bartoli**, et al. "Contextualized Knowledge Graph Embeddings for Activity Prediction in Service Robotics." Workshop on Semantic Scene Understanding for Human-Robot Interaction, ACM/IEEE International Conference on Human Robot Interaction (HRI) 2023.

DEMONSTRATIONS

1. "They Develop Social Robots" Swedish Foundation for Strategic Research, Stockholm 11.2023
2. "NEST-project PerCorSo Spotlight" Wallenberg AI, Autonomous Systems and Software Program, Stockholm 09.2024

TEACHING

KTH Royal Institute of Technology (2023-...)

Hosting labs and lectures, preparing course materials (assignments and lectures), grading, hosting office hours

1. Social Robotics with Iolanda Leite (DD2413)
2. Artificial Intelligence with Jana Tumova (DD2380)

Sapienza University of Rome (2021-2022)

1. Machine Learning with Prof. Luca Iocchi
2. Artificial Intelligence with Prof. Daniele Nardi

AWARDS

1. 2022 – Selected for the Honor Course of my university of the masters degree (5% of the best student in the department)
2. 2021 - Team Leader of Sapienza team "SPQR CocoNuts", winner of Episode 2 of SciRoc Challenge
<https://www.bolognawelcome.com/en/events/fairs/sciroc-2021-smart-city-robotics-challenge-en>
3. 2020 – Selected for the Honor Course of my university of the bachelors degree (5% of the best student in the department)

SERVICE

Reviewer

1. IEEE International Conference on Robotics and Automation (ICRA)
2. IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)
3. ACM/IEEE International Conference of Human-Robot Interaction (HRI)
4. IEEE International Conference on Robot & Human Interactive Communication (RO-MAN)